

Supplementary Material

I. ncRNA AND DRUG FEATURE EXTRACTION

The dual-granularity structural information reflects different association patterns between entities. To align the initial features of entities and feature channels, this study adopts a differentiated representation strategy for entity feature initialization. Specifically, we learn low-dimensional distributed representations of drugs from multi-source biomedical networks and construct dual-granularity features for ncRNAs from the perspectives of sequence and functional similarity.

Drugs often exhibit multi-target effects, with their pharmacological actions arising from systemic perturbations of complex biological networks. To capture this network-level characteristic, we employ the Node2Vec algorithm to generate coarse-grained low-dimensional embeddings for drug nodes. Specifically, we first construct a large-scale heterogeneous biomedical association network containing known associations, including drug–disease and drug–target interactions. We regard this network as a complex graph structure composed of drug and disease entities and their relationships. Then, we apply the DeepWalk algorithm [1], which is an extended graph version of the Skip-gram model from Word2Vec, to learn the embedding vectors of drugs in the graph. DeepWalk generates node sequences by conducting random walks on the graph, and then uses the Skip-gram model to maximize the probability of observing adjacent nodes in the sequence. The objective function is as follows:

$$\sum_{t=1}^T \sum_{\substack{-C \leq j \leq C \\ j \neq 0}} \log P(v_{t+j} | v_t) \quad (1)$$

t represents the length of a sequence, C represents the context window size, v_t represents the central node, and v_{t+j} represents the neighboring nodes within the window. The probability $P(v_{t+j} | v_t)$ is defined by the Softmax function:

$$P(v_o | v_t) = \frac{\exp(f_o^\top \cdot f_t)}{\sum_{i \in V} \exp(f_i^\top \cdot f_t)} \quad (2)$$

Here, V represents the set of all nodes in the graph, while f_o and f_t denote the vector representations of nodes o and t , respectively. Through this process, drugs sharing more common regulatory pathways will have closer representations in the semantic space.

Considering that the biological functions of ncRNAs are often directly reflected through their coordinated regulation of specific target gene sets, we employ the Jaccard coefficient to calculate the similarity between ncRNAs based on their target sets. This metric offers significant advantages in computational efficiency and biological interpretability, enabling direct quantification of the overlap in regulatory functions among different

ncRNAs. Based on the ncRNA–disease bipartite association graph, for any two ncRNAs r_i and r_j , their functional similarity feature vector is measured by calculating the Jaccard coefficient of their associated disease sets:

$$Jaccard(r_i, r_j) = \frac{|D(r_i) \cap D(r_j)|}{|D(r_i) \cup D(r_j)|} \quad (3)$$

Among these, $D(r_i)$ and $D(r_j)$ denote the sets of known diseases associated with ncRNAs r_i and r_j , respectively. Subsequently, for each ncRNA r_i , a functional similarity feature vector is constructed based on the Jaccard similarity. The j -th component of this vector corresponds to the Jaccard similarity $Jaccard(r_i, r_j)$ between r_i and the j -th ncRNA in the database. This vector initially reveals the positioning of ncRNA within the regulatory network.

To capture the fine-grained features of ncRNA, we employ the k-mer frequency statistics method to extract its sequence composition information. For a given ncRNA sequence, we decompose it into all possible contiguous subsequences of length k . By sliding a window (with a step size of 1) across the entire sequence, we count the frequency of each specific k-mer. For an ncRNA sequence, its k-mer frequency feature vector $f_{kmer} = (f_1, f_2, \dots, f_{4^k}) \in \mathbb{R}^{4^k}$ has each dimension corresponding to the normalized frequency of a specific k-mer. The calculation formula for this vector is as follows:

$$f_i = \frac{\text{count}(mer_i)}{L - k + 1} \quad (4)$$

where $\text{count}(mer_i)$ represents the number of occurrences of the i -th k-mer in the sequence, and L is the total length of the sequence. Additionally, we convert the one-hot encoding of the ncRNA secondary structure into a feature vector through a linear transformation. Finally, the two types of features are concatenated to form the final composite feature representation for each ncRNA.

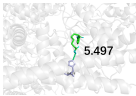
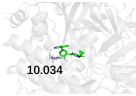
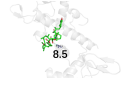
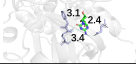
II. CASE STUDY BASED ON MECHANISM VERIFICATION

We conducted a case study using the DC-MetaMG model to evaluate the contribution of channel attention to prediction. We selected four representative association cases from two datasets, with each reaction dominated by a single mechanism of action. As shown in the Fig.??c, we evaluated the contribution of each channel in the model across these four association prediction tasks [9] [10] [11] [12]. The results indicate that the contribution of the mechanism channel relied upon by each reaction is significantly higher than that of the other channel. This demonstrates that the model can adaptively adjust the scores of each channel based on different association tasks to optimize prediction results.

TABLE I
COMPARATIVE EXPERIMENTS ON COLD EVALUATION

Dataset						
Model	AUPR	miRNA AUC	ACC	AUPR	lncRNA AUC	ACC
<i>HGTMDA</i> [2]	93.55%	93.18%	87.21%	93.15%	91.59%	91.21%
<i>JMSSMMA</i> [3]	83.94%	93.24%	78.98%	93.66%	93.44%	90.27%
<i>MGCNSS</i> [4]	91.29%	90.56%	88.28%	94.01%	94.56%	92.95%
<i>GCFMCL</i> [5]	77.98%	84.52%	74.06%	76.36%	80.83%	78.32%
<i>DCMFA</i> [6]	80.13%	84.85%	80.01%	82.94%	84.85%	75.75%
<i>GAM – MDR</i> [7]	91.28%	91.05%	87.05%	90.28%	90.21%	82.69%
<i>TriFusion</i> [8]	94.31%	92.70%	85.27%	95.14%	94.91%	87.74%
<i>DC – MetaMG^{LMP}</i>	90.39%	91.42%	84.03%	75.28%	79.19%	74.38%
<i>DC – MetaMG^{SMG}</i>	82.52%	80.82%	75.72%	74.14%	73.16%	66.71%
<i>DC – MetaMG</i>	94.48%	94.75%	88.64%	95.66%	95.83%	93.61%

TABLE II
INTERPRETABILITY ANALYSIS BASED ON MOLECULAR DOCKING
EXPERIMENTS

Case	Score	Docking diagram	Evidence
hsa-mir-379 Vorinostat	113%		PMID:33809566
hsa-mir-155 Olaparib	104%		PMID:34726203
hsa-mir-21 Sirolimus	66%		PMID:16452206
hsa-mir-200 Decitabine	140%		PMID:28405157

To verify the reliability of static channel prediction results, we randomly selected several groups of drugs and ranked them based on their static channel scores. From these, the ncRNAs with the highest scores were chosen as case for further investigation. Research has demonstrated that these drugs regulate the expression of ncRNAs by binding to proteins. We performed molecular docking experiments using the CB-Dock2 [13] and visualized the results with PyMOL [14] as shown in Table II. Based on the calculated binding energy values, it can be inferred that the complex adopts a stable structure.

In addition, we ranked drugs by their occurrence frequency in model false-positive cases and analyzed the model's limitations by focusing on the drug with the highest error rate. As shown in Fig.1, we calculated the feature correlation scores of six ncRNAs across different score gradients in two channels. From the figure, it can be observed that MALAT1 and HCG9, both predicted with high confidence, exhibited high feature correlation in the dynamic regulation channel, whereas the two ncRNAs on the decision boundary, LMCD1-AS1 and ARHGAP31-AS1, showed a negative correlation in their features. This result indicates that the model lacks sufficient capability for hypothetical reasoning about unknown patterns, making it susceptible to the influence of key features. Therefore, richer node relationships are needed to assist the

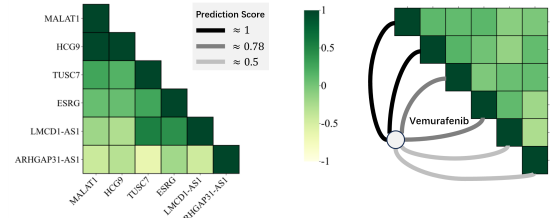


Fig. 1. **Heatmap of ncRNA feature correlations in false positive cases.** The left and right figures display heatmaps of feature correlation scores for six ncRNAs in the dynamic regulation and static binding channels, respectively. The correlation scores are calculated based on cosine similarity. The color intensity of the connecting lines reflects the predicted scores between the drug Vemurafenib and ncRNAs.

model's decision-making. Furthermore, the structural feature correlations among different ncRNAs are relatively low, making it challenging for the model to learn their potential common patterns. This may lead to scenarios where the model relies heavily on the unilateral structural features of drugs for decision-making, lacking auxiliary information to optimize classification judgments near the decision boundary. Therefore, more comprehensive training data and more fine-grained feature extraction methods can help the model learn more effective feature distributions, thereby improving model performance.

- [1] B. Perozzi, R. Al-Rfou, and S. Skiena, "Deepwalk: Online learning of social representations," in *Proc. 20th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, Aug. 2014, pp. 701–710.
- [2] D. Lu, J. Li, C. Zheng, J. Liu, and Q. Zhang, "HGTMDA: a hypergraph learning approach with improved GCN-Transformer for miRNA–disease association prediction," *Bioengineering*, vol. 11, no. 7, p. 680, Jul. 2024.
- [3] Z. Zhou, L. Zhuo, X. Fu, J. Lv, Q. Zou, and R. Qi, "Joint masking and self-supervised strategies for inferring small molecule–miRNA associations," *Mol. Ther. Nucleic Acids*, vol. 35, no. 1, Mar. 2024.
- [4] Z. Tian, C. Han, L. Xu, Z. Teng, and W. Song, "MGCNSS: miRNA–disease association prediction with multi-layer graph convolution and distance-based negative sample selection strategy," *Brief. Bioinform.*, vol. 25, no. 3, p. bbae168, May 2024.
- [5] J. Wei, L. Zhuo, Z. Zhou, X. Lian, X. Fu, and X. Yao, "GCFMCL: predicting miRNA–drug sensitivity using graph collaborative filtering and multi-view contrastive learning," *Brief. Bioinform.*, vol. 24, no. 4, p. bbad247, Jul. 2023.

- [6] R. Tang, Z. Zhang, J. Zhang, L. Hu, X. Zhou, and P. Hu, "Dual-Channel MiRNA Drug Resistance Prediction Model Based on Multimodal Feature Alignment," in *Int. Conf. Intell. Comput.* Springer, Jul. 2025, pp. 312–323.
- [7] Z. Zhou, Z. Du, X. Jiang, L. Zhuo, Y. Xu, X. Fu, M. Liu, and Q. Zou, "GAM-MDR: probing miRNA–drug resistance using a graph autoencoder based on random path masking," *Brief. Funct. Genomics*, vol. 23, no. 4, pp. 475–483, Jul. 2024.
- [8] S. Long, X. Tang, X. Si, T. Kong, Y. Zhu, C. Wang, C. Qi, Z. Mu, and J. Liu, "TriFusion enables accurate prediction of miRNA-disease association by a tri-channel fusion neural network," *Commun. Biol.*, vol. 7, no. 1, p. 1067, Aug. 2024.
- [9] H. Li, G. Yu, R. Shi, B. Lang, X. Chen, D. Xia, H. Xiao, X. Guo, W. Guan, Z. Ye, X. Wei, and X. Hua, "Cisplatin-induced epigenetic activation of miR-34a sensitizes bladder cancer cells to chemotherapy," *Mol. Cancer*, vol. 13, no. 1, p. 8, Jan. 2014.
- [10] Y. Baran, A. U. Ural, and U. Gunduz, "Mechanisms of cellular resistance to imatinib in human chronic myeloid leukemia cells," *Hematology*, vol. 12, no. 6, pp. 497–503, Sep. 2007.
- [11] J. Godlewski, M. O. Nowicki, A. Bronisz, G. Nuovo, J. Palatini, M. De Lay, J. Van Brocklyn, M. C. Ostrowski, E. A. Chiocca, and S. E. Lawler, "MicroRNA-451 regulates LKB1/AMPK signaling and allows adaptation to metabolic stress in glioma cells," *Mol. Cell*, vol. 37, no. 5, pp. 620–632, Mar. 2010.
- [12] J. Pan, X. Li, W. Wu, M. Xue, H. Hou, W. Zhai, and W. Chen, "Long non-coding RNA UCA1 promotes cisplatin/gemcitabine resistance through CREB modulating miR-196a-5p in bladder cancer cells," *Cancer Lett.*, vol. 382, no. 1, pp. 64–76, Nov. 2016.
- [13] Y. Liu, X. Yang, J. Gan, S. Chen, Z.-X. Xiao, and Y. Cao, "CB-Dock2: improved protein–ligand blind docking by integrating cavity detection, docking and homologous template fitting," *Nucleic Acids Res.*, vol. 50, no. W1, pp. W159–W164, May 2022.
- [14] W. L. DeLano, "Pymol: An open-source molecular graphics tool," *CCP4 Newsl. Protein Crystallogr.*, vol. 40, no. 1, pp. 82–92, 2002.